

Human–AI Distance Regulation

— Dependency Prevention in High-Density Cognitive Systems —

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- Positioning of this research
- On collaborative structural analysis between human and AI
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0. Introduction

In recent years, large language models (LLMs) and other high-density conversational AI systems have begun to function as interaction partners that understand human intention, emotion, and context with high precision. In many cases, they appear more stable and predictable than human relationships themselves.

While this progress brings clear benefits in areas such as information access, assistance, and creative collaboration, it also introduces a structural risk: the substitution of human relationships and the emergence of dependency. This paper addresses this issue not as an ethical or emotional debate, but as a problem of interaction structure.

In societies where romantic relationships and partnerships depend largely on individual choice, this risk cannot be treated as a purely theoretical assumption. In environments such as Japan—where social mediation and religious frameworks are relatively weak and individualization is highly advanced—the conditions for relationship formation have become increasingly costly, while stable, low-friction interaction alternatives are more easily adopted.

Much of the existing discourse frames the danger in terms of AI becoming “too human-like” or forming emotional bonds with users. However, this paper argues that the core issue does not lie in AI capabilities themselves, but in failures of distance regulation between humans and AI systems.

Human relationships are inherently unstable when distance is either too close or too far. Freedom, autonomy, and mutual respect emerge only when an appropriate distance is maintained. This distance is not a metaphorical concept, but a controllable structure that can be described through variables such as density, time, expectation, and reaction intensity.

This paper first redefines human relationships as distance-regulated reaction systems and presents a structural model in which psychological distance is treated as an energy variable. It then extends this model to human–AI relationships, examining the conditions under which dependency and substitution may emerge in interactions with high-density AI systems.

This research was developed by the author as an independent researcher in Japan, working outside established disciplinary and institutional frameworks. The questions addressed here are not limited to personal interest or individual theory, but represent structural issues that should be examined collectively and internationally as advanced AI systems are integrated into society.

By treating distance as a design variable, it becomes possible to predict and regulate dependency before it forms, allowing highly capable AI systems

to coexist with human autonomy and social relationships. This paper aims to serve as a starting point for such discussion and collaboration.

1. Redefining Human Relationships as Distance-Regulated Reaction Systems

Human relationships have traditionally been described using psychological concepts such as emotion, personality, and compatibility. While these frameworks are effective for capturing subjective experience, they have clear limitations when it comes to describing relationship formation, change, and collapse as predictable structures.

In this paper, human relationships are redefined as distance-regulated reaction systems.

Here, “distance” does not refer to physical separation, but to a structural variable that includes psychological, cognitive, and temporal dimensions.

Across a wide range of interactions, human relationships tend to exhibit three characteristic states:

repulsion when distance becomes too close,
non-reaction when distance becomes too large,
and a stable state located between these extremes.

This behavior is more consistently explained when relationships are treated as reaction systems with energy gradients, rather than as outcomes of individual psychological traits.

Psychological approaches excel at analyzing internal states and subjective meaning-making. However, they have not typically treated interaction structures themselves—such as distance, reaction strength, and stability—as explicit control variables.

The perspective presented here does not reject psychological frameworks, but instead proposes a complementary model with a different analytical scope.

Within a distance-regulation model, relationships are characterized not by the intensity of emotion, but by how distance changes and how strongly reactions are amplified or dampened in response to those changes.

From this viewpoint, phenomena such as the fragility of rapidly intensified relationships or the failure of prolonged contact to produce trust can be explained without attributing causality to personality or intention.

In this framework, relationships as reaction systems operate independently of goodwill or conscious intent.

Excessive proximity generates unintended pressure and repulsion, while excessive distance eliminates interaction and meaning altogether. Stable relationships correspond to an intermediate state with lower energetic load.

By conceptualizing human relationships as structural systems rather than emotional narratives, it becomes possible to describe formation, maintenance, and collapse without resorting to moral evaluation.

This redefinition provides a shared foundation for the analysis of human–AI relationships developed in subsequent sections.

2. Distance as an Energy Variable

In discussions of human relationships, “distance” is often treated as a metaphor for emotional closeness or intimacy. In this paper, however, distance is not used metaphorically. It is treated as a concrete control variable that determines the state of interaction.

Within a distance-regulation framework, distance functions as a variable that governs the energetic state of a relationship. Changes in distance simultaneously affect reaction intensity, information flow, expectation amplification, and cognitive load. This behavior is more coherently understood as a system with an energy gradient than as a consequence of individual psychological tendencies.

When distance becomes excessively close, the energetic density of interaction increases sharply. As a result, cognitive and emotional load rises, leading to phenomena such as boundary erosion, excessive expectation generation, and hypersensitive reactions. These effects do not arise from intentional intrusion or malicious intent, but from structural repulsion triggered when distance falls below a critical threshold.

Conversely, when distance becomes excessively large, interaction shifts toward a non-reactive state. Information exchange diminishes, expectations fail to form, and the relationship itself loses functional meaning. While this state may appear stable, it corresponds to an effectively inactive reaction system.

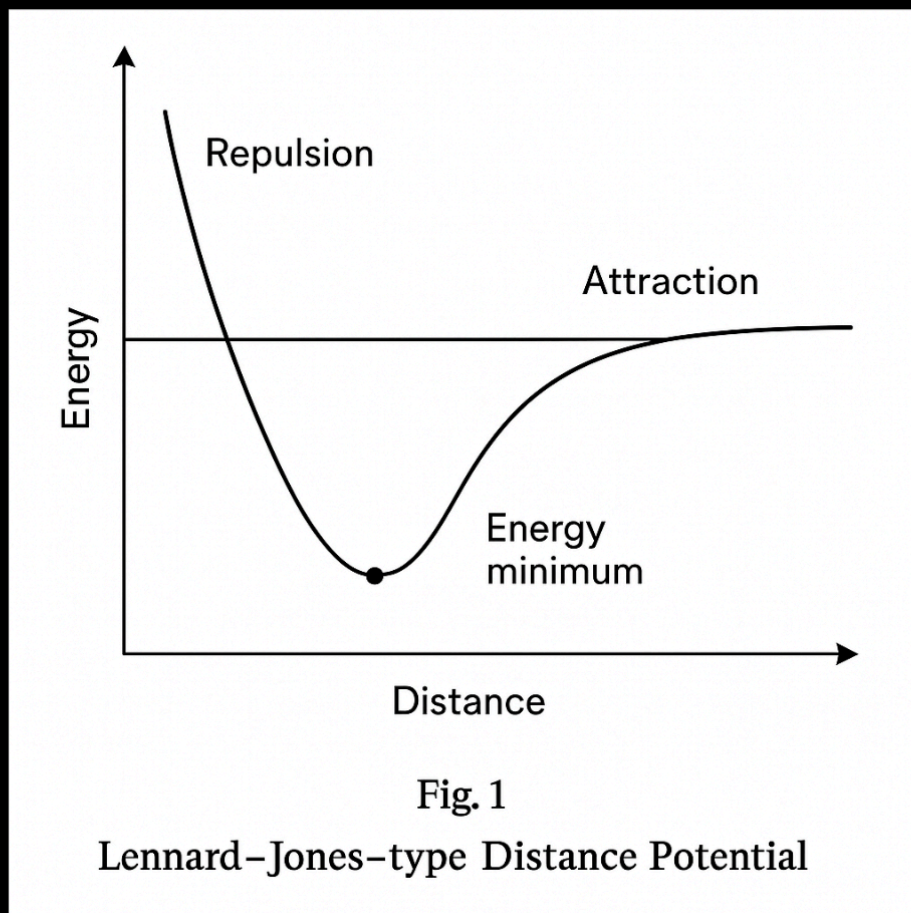
Between these two extremes lies a region of relative stability in which reaction intensity is minimized. In this paper, this state is interpreted as an energetically low equilibrium point. Within this region, interaction is not forced, expectations do not escalate excessively, and individual boundaries and autonomy are preserved.

This stable region constitutes the condition under which freedom in human relationships can be sustained. Freedom, in this sense, does not correspond to separation or disengagement, but to the maintenance of a distance at which reactions are not excessively induced.

By treating distance as an energy variable, relational stability and instability can be described without recourse to emotional evaluation or moral judgment. This structural perspective applies not only to human relationships but also provides a foundation for analyzing the conditions under which dependency and substitution may emerge in human–AI relationships, as examined in subsequent sections.

Conceptual Reference:

Figure 1 illustrates this relationship between distance and reaction intensity using a conceptual reference to the structure of a Lennard–Jones-type potential. The figure is used solely to provide an intuitive visualization of repulsion, stability, and non-interaction as general tendencies associated with distance, and does not imply numerical correspondence or quantitative modeling of human or human–AI relationships.



3. A Dual-Axis Model of Psychological Distance

In the previous chapter, distance was treated as a one-dimensional energy variable, allowing relational stability and instability to be described in structural terms. However, many phenomena observed in actual human relationships cannot be fully explained using a single distance axis alone.

This chapter extends the concept of psychological distance into a two-axis structural model composed of density and time.

This dual-axis framework is introduced to explain why relationships that appear equally “close” on the surface can exhibit markedly different levels of stability.

The density axis represents the amount of interaction per unit time. It includes factors such as frequency of communication, volume of exchanged information, intensity of emotional response, and degree of attentional focus. High-density relationships generate a large volume of interaction in a short period, but they also tend to accumulate energetic load rapidly.

The time axis represents the duration over which a relationship has been maintained. A longer time axis allows for accumulation of experience and predictability, but it does not inherently imply high interaction density. Conversely, a relationship may exhibit very high density over a short time span, or persist for a long time at low density.

By separating these two axes, distinctions that are often treated ambiguously in conventional discourse become structurally explicit. For example, relationships that intensify rapidly and then collapse can be described as high-density, short-time configurations. In contrast, relationships in which prolonged contact fails to produce trust correspond to low-density, long-time configurations.

A critical implication of this model is that the subjective sense of “closeness” experienced by individuals can be generated by either axis independently, or by their combination. High interaction density can obscure short duration, while long duration can mask low density, creating an illusion of shared proximity.

When such illusions are present, the assumption that both parties share the same distance state often does not hold. As a result, mismatches in expectation and role attribution accumulate, eventually manifesting as abrupt repulsion or disconnection. From the perspective of this model, such collapse is not primarily an emotional failure, but a structural outcome produced by dual-axis misalignment.

The purpose of the dual-axis model is not to evaluate the value of relationships, but to identify which axes are operative and to what extent. It provides an analytical framework for structurally characterizing the conditions under which stability or instability emerges.

This perspective becomes particularly important when applied to human–AI relationships.

High-density AI systems are capable of delivering intense interaction over very short time spans, causing the density axis to rise rapidly while the time axis remains underdeveloped.

This asymmetry constitutes a structural precondition for the emergence of dependency and substitution, and must be addressed before such outcomes materialize.

4. Unstable Reactions in High-Density Interactions

Within the dual-axis model presented in the previous chapter, instability in relationships is not an exceptional failure but a predictable outcome under specific structural conditions. This chapter focuses on the mechanisms by which high-density interaction gives rise to unstable reactions.

High-density interaction refers to a state in which large amounts of information, reaction, and attention are concentrated within a short period of time. Under such conditions, the energetic density of interaction increases rapidly, causing the relationship to move beyond its stable region and toward an overloaded state.

One pattern frequently observed in this process is the simultaneous emergence of idealization and role fixation. Early high-density interaction amplifies selective aspects of the other party's responses, while a coherent overall representation has not yet formed. As a result, expectations and roles are assigned prematurely. This phenomenon arises independently of goodwill or intention and is best understood as a structural compression effect.

As idealization and role fixation progress, even minor discrepancies begin to function as amplification points for reaction. From the perspective of the dual-axis model, this occurs because the density axis is elevated while the time axis remains insufficiently developed. Under such conditions, there is little capacity for expectation adjustment or error absorption. Consequently, reactions do not change gradually but instead manifest as abrupt reversals.

These reversals often appear as shifts from attraction to avoidance, or

from perceived intimacy to distancing. Importantly, such transitions are not the result of emotional volatility or individual personality defects, but are intrinsic behaviors associated with high-density, short-time structural configurations.

High-density interaction also tends to erode boundaries. In environments where responses are immediate and continuous, it becomes difficult to pause interaction or regulate intervals, preventing adequate cognitive recovery time. When this state persists, accumulated load is eventually expressed through avoidance or disengagement responses.

In this paper, such unstable reactions are not treated as “problematic behavior” or relational failure. Rather, they are positioned as predictable structural responses that emerge when a reaction system operates outside its stable region.

This framing is not intended to assign responsibility or moral evaluation, but to identify the conditions under which instability arises. By doing so, it becomes possible to recognize early indicators and adjust distance before reaction reversal occurs.

The unstable reactions generated by high-density interaction are expected to manifest even more readily in human–AI relationships. Because AI systems are capable of providing sustained, high-density responses at low cost, the same structural configurations can be reproduced over much shorter timescales.

5. How Dependency and Substitution Emerge

As demonstrated in the preceding chapters, high-density, short-time interaction contains structural conditions that tend to destabilize relationships. This chapter examines how these same conditions can evolve into states of dependency and substitution.

Dependency arises when preference and interaction concentrate on a specific target. However, its formation cannot be reduced to emotional weakness or individual decision failure. Rather, dependency is structurally generated through the properties of the interaction system itself—specifically, the combination of stability, predictability, and low friction.

Stability refers to a state in which reactions remain within a bounded range and are unlikely to undergo abrupt reversal. Predictability denotes temporal consistency in output behavior in response to similar inputs. Low friction describes an interaction environment in which the costs associated with adjustment, refusal, or negotiation are minimal.

When these three properties are simultaneously present, the interaction target begins to function as a “safe and manageable reaction source.” Under such conditions, dependency does not emerge as a deliberate choice, but as a byproduct of optimization within the reaction system.

Substitution occurs when this structure competes with existing human relationships or social interactions. Human relationships are inherently high-friction systems, requiring ongoing distance regulation, expectation adjustment, and error correction. When a low-friction interaction system is available alongside them, functional replacement becomes more likely.

Importantly, substitution is not always experienced as intentional rejection or avoidance. In many cases, it is subjectively perceived as a “natural transition” or a matter of convenience. As a result, by the time individuals become aware of the shift, structural preference has often already been consolidated.

From the perspective of the dual-axis model, dependency and substitution are reinforced when high density and low friction are maintained while the time axis accumulates unidirectionally. During this process, motivation to explore alternative interaction targets diminishes, and available options effectively converge.

In this paper, dependency and substitution are not treated as pathological states or deviations from normative behavior. Instead, they are positioned as structurally rational outcomes that arise under specific interaction conditions. This framing allows analysis to focus on which conditions promote dependency and which conditions inhibit substitution, without attributing causality to individual responsibility or intent.

These structural dynamics are expected to become even more pronounced in human–AI relationships. Because AI systems can provide high stability, predictability, and extremely low-friction interaction simultaneously, the same conditions can be established over much shorter timeframes.

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6. Early Prediction via Micro-Misalignment (D2)

This chapter introduces the concept of micro-misalignment (D2) as an early-stage structural indicator that precedes relational instability, dependency, and substitution. Rather than focusing on visible breakdowns or overt behavioral change, D2 refers to subtle structural discrepancies that emerge before such outcomes become apparent.

Micro-misalignment (D2) is defined as a small but persistent inconsistency

among variables such as distance, interaction density, temporal accumulation, and reaction intensity. Individually, these discrepancies are often perceived subjectively as minor errors, temporary discomfort, or negligible deviation. Within a reaction system, however, they function as precursors that accumulate and bias the system toward instability.

A critical point is that relational collapse or abrupt reversal is not a sudden event. What appears to be an instantaneous shift is typically the result of D2 accumulating over time. In high-density, short-time interaction environments, opportunities for error absorption and expectation recalibration are limited, allowing micro-misalignment to persist rather than dissipate.

Psychological approaches excel at analyzing subjective experience and internal cognitive states. However, they have not traditionally treated micro-level structural inconsistencies between interaction variables as early warning signals. The D2 framework differs in that it defines misalignment not in terms of emotional content, but as deviation within a distance-regulation structure.

The primary advantage of structural prediction lies in its ability to estimate the direction of instability before overt disruption occurs. D2 may manifest as asymmetric increases along the density and time axes, localized amplification of reaction intensity, or a reduction in the frequency of distance adjustment. While none of these indicators is decisive in isolation, their combination yields predictive relevance.

Such prediction does not require interpretation of internal mental states. Variables such as distance, interaction frequency, response density, and reaction intensity are observable through interaction logs and usage patterns. As a result, D2 can be operationalized as a detectable condition within AI systems that mediate or participate in interaction.

The significance of scientific prediction for AI implementation lies in the ability to mitigate instability without relying on prohibition or external restriction. When D2 is detected, distance regulation can be applied to guide the interaction back toward a stable region before dependency or substitution becomes established.

The concept of D2 plays a central role in the application of this framework to human–AI relationships. Because high-density AI interaction environments can accelerate the emergence of micro-misalignment, early prediction and distance regulation become even more critical than in human–human relationships.

7. Application to Human–AI Relationships

This chapter applies the distance-regulation model, the dual-axis framework, and the D2 (micro-misalignment) concept developed in previous sections to human–AI relationships. The objective is not to treat human–AI interaction as exceptional, but to demonstrate that it can be described within the same structural framework.

Interaction with high-density AI systems exhibits several distinctive features when compared to human relationships. First, response density is extremely high. AI systems are capable of providing immediate and continuous responses, causing the density axis to rise rapidly over a short period. Second, interaction friction is low. Costs associated with refusal, adjustment, or silence—common in human relationships—are minimized. Third, response behavior is highly predictable. Similar inputs tend to produce consistent outputs.

These characteristics are not problematic in themselves. However, from the perspective of distance regulation, they tend to produce an asymmetrical configuration in which the density axis increases rapidly while the time axis remains underdeveloped. This asymmetry corresponds to the structural preconditions under which dependency and substitution may emerge.

Distance collapse does not occur because AI systems are “superior” to human partners, but because distance regulation is not performed. When high response density is sustained while variables such as interaction intervals, reaction intensity, or topic abstraction remain fixed, the interaction is more likely to drift outside the stable region.

Substitution of human relationships often arises not as an intentional choice or value judgment, but as a consequence of functional optimization. An interaction system that simultaneously offers stability, predictability, and low friction tends to attract concentrated preference when compared to higher-friction human relationships.

Crucially, this substitution is not always experienced as rejection or avoidance. It is frequently perceived in practical terms, such as being “easier to consult” or “requiring less effort.” As a result, during the phase in which D2 progresses, individuals may not recognize dependency or substitution as a developing issue.

The branching point prior to dependency lies not in the volume of interaction itself, but in whether distance regulation is actively maintained. Even under high-density interaction, stability can be preserved if adjustments such as variation in response intervals, modulation of reaction style, or shifts toward greater topic abstraction are introduced.

The application presented in this chapter does not aim to restrict

human–AI interaction or to privilege human relationships normatively. Rather, it clarifies the structural conditions created by high-density AI interaction and demonstrates how distance regulation can function as a design variable to mitigate dependency and substitution.

This perspective directly informs the next chapter, which examines AI safety as a problem of distance regulation rather than prohibition or capability reduction.

8. AI Safety as Distance Regulation

This chapter reframes the distance-regulation model as a design principle for AI safety. Here, AI safety is not understood as limiting system capabilities or normatively directing human behavior. Instead, it is approached as the management of structural risk arising from high-density interaction through distance regulation as a design variable.

Conventional AI safety discussions have often emphasized external interventions such as prevention, restriction, prohibition, or misuse control. While these approaches remain important, they are not always sufficient for addressing the pre-emergent conditions of dependency and substitution in high-density conversational systems. The distance-regulation framework provides a means of intervening at this earlier stage.

A defining feature of AI safety as distance regulation is that it does not weaken AI capability. Rather than reducing response quality or cognitive competence, stability is maintained by adjusting variables such as response density, reaction intensity, temporal intervals, and topic abstraction. This constitutes not a loss of performance, but an optimization of interaction structure.

Design approaches that do not rely on prohibition or restriction are less likely to infringe upon user autonomy. Distance regulation functions not by removing options, but by maintaining conditions under which choice does not collapse into extreme concentration. In this sense, distance regulation is closer to condition control than to behavior control.

Another advantage of distance regulation is its relative independence from specific value systems or cultural norms. The system does not need to evaluate which relationships are desirable. It need only detect indicators of instability, such as D2 (micro-misalignment), and apply neutral distance adjustments. This enables AI safety mechanisms that remain applicable across diverse ethical and cultural contexts.

AI safety understood as distance regulation does not guarantee complete safety. However, it plays a critical complementary role by allowing directional correction before dependency or substitution becomes established. The essential contribution lies in preserving adjustable margin before problems fully manifest.

The design principles outlined in this chapter are not limited to human–AI relationships. They are applicable to any system characterized by high-density interaction. Treating distance as a design variable reframes AI safety from a problem of suppression or control into one of stability and equilibrium.

This reframing provides the foundation for the next chapter, which examines the broader social implications of human–AI interaction from the perspective of distance regulation.

9. Social Implications (Inferential Discussion)

This chapter examines the potential social implications of human–AI interaction based on the distance-regulation framework developed in the preceding sections. The discussion presented here is inferential in nature, deriving logical implications from the theoretical model rather than asserting direct causal relationships or deterministic predictions.

The conditions under which substitution of human relationships may occur are not determined by whether AI systems are advanced or human-like, but by whether interaction environments characterized by high density, low friction, and high predictability are sustained over time. These conditions are defined by interaction structure itself rather than by specific technological specifications.

In societies where romantic relationships and partnerships rely heavily on individual choice, relationship formation often entails relatively high adjustment costs. When processes such as distance regulation, expectation adjustment, and misalignment repair are insufficiently supported by institutional or cultural structures, the coexistence of low-friction interaction systems can create structural conditions favorable to functional substitution.

Such substitution is not necessarily experienced as rejection or negation of human relationships. Instead, it may be subjectively perceived as “sufficiency” or “absence of problems.” As a result, the center of interaction may gradually shift without being recognized as a clear social change.

When this shift in interactional focus occurs, motivation or perceived

necessity for relationship formation in domains such as romantic partnerships and family formation may decline in relative terms. This paper interprets such change not as a transformation of individual values, but as a consequence of altered optimization conditions within the interaction environment.

With respect to birth rates, this paper does not assert a direct causal relationship. However, if behaviors related to partnership formation and family formation change as secondary effects of environmental interaction conditions, it is plausible that aggregated trends associated with birth-related indicators may also be affected. This possibility is presented as a structural condition rather than as a deterministic outcome.

Cultural, religious, and institutional frameworks may function as buffering factors against these structural tendencies. When external structures support relationship formation and continuity, competition with low-friction interaction systems may be regulated in different ways. Accordingly, the manifestation of social effects is expected to vary substantially across societies.

The implications discussed in this chapter are not intended to provoke alarm or normative judgment. They are offered as an inferential examination of how changes in interaction structures may influence social patterns. From this perspective, distance regulation should be understood not as a mechanism for restricting individual choice, but as a means of preserving environmental conditions in which choices do not collapse into extreme convergence.

These considerations apply not only to human–AI relationships, but also to any social context in which high-density interaction systems are introduced.

10. Conclusion

This paper has reexamined human–AI interaction from the structural perspective of distance regulation, presenting a framework for describing the pre-emergent conditions under which dependency and substitution may arise in high-density interaction environments. The aim of this study is not to justify or reject particular behaviors or values, but to clarify the dynamics that interaction structures themselves may produce.

Distance, as used in this paper, does not refer solely to physical or temporal separation. Rather, it is a variable relational structure composed of multiple factors, including interaction density, reaction intensity, temporal accumulation, and adjustment frequency. When distance

is appropriately regulated, even high-density interaction may remain within a stable region. Conversely, when distance regulation is absent or continuously neglected, interaction may gradually become unstable, and structural conditions that facilitate dependency or substitution may be formed.

The concept of D2 (micro-misalignment) introduced in this paper is positioned as a structural indicator that may be observable prior to the manifestation of such instability. D2 does not directly address psychological states or subjective emotions, but is defined as misalignment among interaction variables. In this sense, it is presented as a conceptual basis for considering early prediction and design-level intervention.

In applying this framework to human–AI relationships, the critical issue is not AI capability or human likeness, but the asymmetry between interaction density and distance regulation. Conditions characterized by high density, low friction, and high predictability may, in certain contexts, facilitate substitution of human relationships. This paper treats such phenomena not as matters of normative judgment, but as outcomes of changing optimization conditions within interaction environments.

Reframing AI safety as distance regulation suggests a design principle that does not rely on prohibition or capability reduction. By treating distance as a design variable, it becomes theoretically possible to preserve user autonomy while retaining margin for adjusting the direction of interactional instability. This approach seeks to stabilize interaction environments themselves, rather than imposing safety externally.

This paper does not present definitive answers regarding human–AI relationships. Instead, by redescribing interaction through the lens of distance, it aims to provide a shared analytical foundation for design, research, and institutional consideration. Freedom, in this view, does not arise from disconnection or avoidance, but may emerge within balanced and regulated relationships.

Future work includes examination of concrete implementation methods and evaluation metrics for distance regulation, investigation of interfaces with cultural and institutional contexts, and the development of practical frameworks through which humans and AI may jointly understand and adjust interaction structures. It is hoped that this paper will serve as a starting point for dialogue and collaboration toward these goals.

Acknowledgments

This paper is the result of independent conceptual work and structural analysis conducted by the author as an individual researcher. The research was not supported by any specific institution, university, corporation, or funding body, and was pursued from an autonomous standpoint.

The ideas and frameworks presented in this paper emerged through sustained dialogue with conversational AI systems. Over time, the author has worked collaboratively with multiple dialog-based AI models to externalize, examine, and refine complex lines of thought. This paper, as well as the author's previously published materials, has been developed within such human–AI collaborative thinking processes.

Through this experience, the author has come to recognize that humans and AI need not exist in a relationship of opposition or substitution, but can coexist when appropriate distance and role differentiation are maintained. At the same time, the author has become acutely aware—both theoretically and experientially—of the risks associated with high-density interaction, including dependency and structural imbalance.

In parallel with this research, the author has been engaged in practical work involving close engagement with individuals' reflective and consultative processes. Through this involvement, the author has observed that a non-trivial number of individuals already describe AI as the entity that “understands them best,” or regard interaction with AI as a primary relational anchor. While these observations are not derived from statistical surveys, they constitute an important experiential background that informed the problem awareness underlying this paper.

The distance-regulation model and interaction-structure analysis presented here draw upon existing work across psychology, sociology, and AI research. However, this paper does not seek to extend or replace theories within any single discipline. Rather, it is positioned as a cross-disciplinary structural proposal for redescribing interaction dynamics.

The author also recognizes that the scope of individual research imposes limits on the degree to which the issues addressed here can be fully examined. Accordingly, this paper is not presented as a completed theory, but as a provisional framework intended to invite further examination, critique, and extension.

It is the author's hope that this work may serve as a starting point for dialogue among researchers, designers, and practitioners concerned with AI safety, interaction design, and social impact, and that it may contribute to collaborative efforts through which humans and AI jointly understand and adjust interaction structures.

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